BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 18 . MODULE 15 OF 17

Module 15: Digestive System

Topics: GI Tract Anatomy and Motility, Digestion and Absorption

TOPIC 1 OF 2 . WEEK 18 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

GI Tract Anatomy and Motility

From mouth to anus, the layers of the gut wall, and how food moves.

The Science

The GI tract is a single tube from mouth to anus, about 9 meters long. Path: mouth, pharynx, esophagus, stomach, small intestine (duodenum, jejunum, ileum), large intestine (cecum, ascending, transverse, descending, sigmoid colon), rectum, anus. Accessory organs (salivary glands, liver, gallbladder, pancreas) deliver enzymes and bile into the tube.

Wall layers (deep to superficial): mucosa (epithelium, lamina propria, muscularis mucosae), submucosa (vessels, submucosal/Meissner plexus), muscularis externa (inner circular and outer longitudinal smooth muscle layers with myenteric/Auerbach plexus between), serosa or adventitia. The two enteric nerve plexuses make up the enteric nervous system, sometimes called the 'second brain' for its independent function.

Motility comes in two flavors. Peristalsis: a wave of smooth muscle contraction that propels content forward. Segmentation: localized contractions that mix without net forward motion (small intestine). Sphincters divide the tract into functional compartments: lower esophageal (cardia), pyloric (stomach to duodenum), ileocecal (ileum to cecum), internal and external anal.

Teaching This Topic

Before the video (drawing prompt)

Have students trace the GI tract from mouth to anus, naming every segment in order, then label the major sphincters.

Common misconceptions to address

- Students think the small intestine is small. It is small in diameter but very long (about 6 meters); the large intestine is shorter (about 1.5 m) but wider.
- Peristalsis and segmentation are sometimes mixed. Peristalsis moves food forward; segmentation mixes it in place.
- Sphincters are thought of as decorative. Each is functionally critical; failure produces specific clinical syndromes (achalasia, gastroparesis, anal incontinence).

Order of operations (what to teach first)

Organ sequence, then the wall layer structure (same throughout, with regional modifications), then motility and sphincters.

Self-test prompts

- Name the three parts of the small intestine in order.
- What is the function of the pyloric sphincter?
- How does peristalsis differ from segmentation?

Why This Matters to Your Patients

Achalasia: the lower esophageal sphincter cannot relax; food and liquid accumulate in the lower esophagus, producing dysphagia and regurgitation. Gastroparesis: delayed stomach emptying, often in diabetics from autonomic neuropathy. Small bowel obstruction: peristalsis pushes against a blockage, causing cramping, distension, vomiting. C. difficile colitis: pseudomembranous colon inflammation after antibiotics; severe diarrhea, electrolyte disturbance. Hirschsprung disease: absence of enteric plexus in part of the colon; obstruction in newborns. Every GI symptom should make you ask which segment is failing and how.

TOPIC 2 OF 2 . WEEK 18 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Digestion and Absorption

Where each macromolecule gets broken down and how absorption happens.

The Science

Each macromolecule has a digestion story. Carbohydrates: salivary amylase begins starch breakdown in the mouth (stops in the stomach because of acid). Pancreatic amylase resumes in the small intestine, producing disaccharides. Brush-border enzymes (maltase, sucrase, lactase) finish the job to monosaccharides, which are absorbed by SGLT1 (Na-glucose symporter) and GLUT2.

Proteins: pepsin in the stomach (activated from pepsinogen by acid) begins. Pancreatic proteases (trypsin, chymotrypsin, carboxypeptidase, activated by enterokinase in the duodenum) continue. Brush-border peptidases finish to amino acids and small peptides, absorbed by amino-acid transporters.

Lipids: bile (made by liver, stored in gallbladder, released into duodenum in response to CCK) emulsifies fats. Pancreatic lipase hydrolyzes triglycerides to monoglycerides and fatty acids. Products form micelles, diffuse into enterocytes, reassemble into chylomicrons, and enter lymph via lacteals. Fat-soluble vitamins (A, D, E, K) ride along.

Absorptive surface: plicae circulares (folds), villi (finger-like projections), microvilli (brush border). Together they multiply surface area roughly 600-fold compared to a smooth tube of the same length. Small intestine = absorption central. Large intestine reclaims water and electrolytes, ferments residual carbohydrates, and forms feces.

Teaching This Topic

Before the video (drawing prompt)

Have students predict where each macromolecule (carbs, proteins, fats) is digested and where it is absorbed. They have to commit first.

Common misconceptions to address

- Students think the stomach does most of the digestion. The stomach starts protein digestion and stores food; the small intestine does the lion's share.
- Fats are thought to enter the blood like carbs and protein. They enter lymph first (chylomicrons via lacteals).
- Bile is sometimes confused with a digestive enzyme. It is not an enzyme; it is an emulsifier that lets enzymes (lipase) access the substrate.

Order of operations (what to teach first)

Each macromolecule's digestion and absorption in turn, then the absorptive surface design as the unifying answer to 'how does so much get absorbed so fast.'

Self-test prompts

- Where is most absorption accomplished?
- What does bile do, and where does it come from?
- Why do fats enter lymph rather than blood directly?

Why This Matters to Your Patients

Lactose intolerance: lactase deficiency; lactose is not digested, ferments in the colon, draws water in osmotically, causing gas and diarrhea. Celiac disease: gluten triggers autoimmune destruction of small intestinal villi; absorption fails; the patient develops nutritional deficiencies. Cystic fibrosis affects the pancreas: thick secretions block enzyme delivery, causing maldigestion. Cholecystectomy (gallbladder removal): bile flow continues but is no longer concentrated and stored; patients often tolerate small fatty meals but not large ones. Bariatric surgery alters absorption and digestion deliberately for weight loss; lifelong vitamin and protein monitoring is essential. GI complaints should prompt thinking about which step of digestion or absorption is failing.